

Digital consumer engagement: examining the impact of native advertising exposure in a social network

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Abstract

Purpose – The digital environment afforded by social networks has created an opportunity to understand more clearly the impact of social media native advertising on advertising processing outcomes. Thus, the current study integrates native advertising with engagement literature to compare engagement outcomes between feed and banner placements before analyzing engagement outcomes of sponsored social media posts by advertising objective. This work aims to contribute to advertising effectiveness literature arguing for the importance of engagement as a measure of effectiveness.

Design/methodology/approach – Facebook advertising data were collected from a convenience sample of 10 Facebook advertisers that accounted for roughly \$414,000 in advertising spend. Panel data, which are also called longitudinal or cross-sectional time-series data, used 26 months of data from the 10 advertisers to measure relationships between native advertising exposure and digital consumer engagement with advertising by advertising objectives of brand awareness, link clicks, conversions, post-engagement and video views.

Findings – Exposure to native advertising was a strong predictor of advertising processing and consumption using the three variables of interest: clicks, comments and shares. Ads reaching consumers while natively consuming content in their feed resulted in statistically significant improvements in impressions and clicks when compared to banner ads. Exposure to native ads was significantly related to all engagement outcomes of interest, except for advertisers who chose post-engagement as their advertising objective.

Practical implications – The results suggest that for advertisers seeking clicks, post-engagement objectives should likely be avoided. For this group, impressions were not related to link clicks but were related to comments and shares. Native advertising placements in the feed, however, are generally more effective than banner ads on Facebook for advertisers seeking engagement.

Research limitations/implications – This research is one of few studies to use longitudinal advertising data to explore engagement effects using real-world data collected from a diverse set of Facebook advertisers over a 26-month period. This study shows that interactive marketers using a social media feed to reach consumers can expect positive outcomes in advertising consumption, affective and cognitive processing and advocacy, but those outcomes may vary by advertising objective.

Originality/value – Given the uniqueness of the data set, the findings contribute to native advertising literature and to the literature on digital consumer engagement with advertising in social media. The study also provides empirical support for the efficacy of native advertising.

Keywords Consumer engagement, Native advertising, Facebook, Panel data, Social media advertising, Sponsored posts

Paper type Research paper



Social networks offer entirely new ecosystems where interactive marketers can engender an organic consumer–brand relationship (Wang, 2021). As such, the feed enables a touch point for marketers to reach desired audiences in a manner that is relatively natural to consumers, subsequently taking advantage of the credibility of the content publisher. Termed native

advertising, these feed placements are characterized as a paid advertisement “that takes the specific form and appearance of editorial content from the publisher itself” (Wojdyski and Evans, 2016, p. 157). In these scenarios, the paid messages are so cohesive that viewers simply feel the message belongs as part of the editorial content (Marks *et al.*, 2019). Over time, by mimicking the site where it appears, marketers may be able to mitigate negative outcomes of disruption that are conventionally associated with online advertising by using native advertising (Campbell and Marks, 2015). Given the practical importance of native advertising, marketers and practitioners can benefit from native advertising.

The tracking of ad exposure in digital environments creates an opportunity for native advertising impact to be uniquely assessed from other environments (Marks *et al.*, 2019). At the same time, engagement outcomes are increasingly becoming a priority for marketers who recognize that engagement is more closely aligned with the long-lasting benefits of relationship marketing (Šonková and Grabowska, 2015). While best practices continue to be tested in the literature, companies are still seeking to understand the meaning behind these interactions (Schultz and Peltier, 2013; Naylor *et al.*, 2012). Additionally, few studies use real-world data to examine the outcomes of native advertising placements in social networks alongside engagement outcomes of advertising processing levels. As such, interactive marketers stand the chance to be more strategic in how they position themselves to build relationships through engagement with their consumers using native advertising in a social network.

Digital consumer engagement with advertising has been defined as how consumers interact, respond and attend to an advertisement in the social media environment. It is posited that native advertising may lead to more positive outcomes in engagement. Past studies have assessed native advertising effectiveness in social media using engagement outcomes in silos, for instance examining sharing behavior (Lee *et al.*, 2016) or conversion (Harms *et al.*, 2017). In addressing these literature gaps, the current work extends Gavilanes and his colleagues’ (2018) model of engagement behavior alongside ad processing to examine native advertising effects by advertising objectives together with dependent variables of clicks, comments and shares. By delineating boundary conditions with specific audience groups (Lee *et al.*, 2016), marketers can become more strategic in how they approach their media placements and plan with specific, desired downstream behaviors in mind. Thus, real-world data from 10 companies and their Facebook advertising analytics were used to compare and examine digital consumer engagement with advertising. The advertisers consist of a variety of consumer product goods, services and business-to-business products. The total spend for this group of advertisers during the 26-month time series was \$413,817.20. The 10 advertisers’ data were used grouped as panels, or cross-sections, to run panel data regressions and explore engagement effects.

Literature review and theoretical conceptualization

Native advertising in social media

Wojdyski and Golan (2016) describe native advertising as, “the practice by which a marketer borrows from the credibility of a content publisher by presenting paid content with a format and location that matches the publisher’s original content,” and that it “has been the primary driving engine of the Internet marketing economy for several years” (p. 1,403). Because an advertisement can be presented in a location and format similar to the publisher’s content, social networks offer an attractive channel for advertisers to market their products. As such, native advertisers benefit from reaching users in their feeds where they may be more likely to engage with an ad. This could be in part due to paid advertising being presented in an increasingly varied format (Campbell *et al.*, 2014) or because it is becoming more difficult to identify (Mitra *et al.*, 2008).

This outstanding empirical question tends to fall into one of two buckets in the native advertising literature (Wojdynski, 2016). The first seeks to understand the circumstances where native advertising is more effective than traditional display advertising in driving consumer–brand engagement, whereas the second stream of literature seeks to understand whether the effectiveness of the native advertisement is a result of the consumer being less aware of the message being a paid advertisement (Wojdynski, 2016). Apart from concern for privacy and ad irritation, native ads have been shown to have a positive relationship with ad avoidance (Baek and Morimoto, 2012), which may be less of a concern particularly on social media.

Wojdynski (2016) identifies three categories of native advertising based on characteristics like the format of the content and the location where the ad is sponsored. The first type he categorized was a sponsored content that is presented in an article-style format, the second was sponsored listings like those found in a Google search and the third was sponsored posts like those found in a social media feed. Harms *et al.* (2019) examined how consumers perceive differences in message intent between article-style native ads and banner ads in an online experiment. The authors found that for 303 respondents, consumers had more positive evaluations of banner-style ads than they did of the article-style native ads which could be related to the level of deception triggered by the advertisement.

Categorizing native advertisements is useful because consumers are in unique motivational states depending on the type and place where they encounter the ad (Marks *et al.*, 2019). For instance, social sharing is not inherent to native advertising (Lieb *et al.*, 2013), but the engagements afforded to a social media user create opportunities for advertisers to achieve desired engagement outcomes (see Figure 1). Additionally, people with strong information-seeking motivation exhibit positive attitudes toward and behavioral responses to native advertising (Lee *et al.*, 2016). Overall, this suggests that if



Figure 1.
A native
advertisement in the
social media feed

an advert is correctly targeted to a consumer who is more likely to seek out the product or service, the resulting effects will be positive.

Regarding the second stream of native advertising literature, researchers have explored the negative effects of native advertisements. If a consumer is not aware of the advertisement and then becomes aware, negative reactions can occur. Investigating the potential for these boomerang effects, [Lee et al. \(2016\)](#) surveyed 550 respondents and found that consumers who perceive native ads as less intrusive were more likely to *share* the ads and suggest this will in turn lead to higher consumption. Additionally, eye-tracking researchers have shown that native ads that are placed in the feed perform better than Facebook's banner placements ([Jung and Heo, 2019](#)). Based on previous literature, it is hypothesized that native advertising will result in better engagement outcomes when compared to Facebook's banner advertisements.

Taken together, the following hypotheses provide a baseline for comparing native advertising with banner advertising:

- H1a.* Native advertisements will result in more impressions when compared to banner advertisements.
- H1b.* Native advertisements will result in more clicks when compared to banner advertisements.

Engagement as a measure of effectiveness

Consumer engagement is becoming more frequently listed as a marketing goal ([Shahbaznezhad and Rashidirad, 2021](#); [Lee et al., 2018](#)), likely due to its ability to help marketers understand how consumers interact, respond and attend to advertising ([Liu-Thompkins et al., 2020](#)). Engagement outcomes are examined in the literature by a variety of media factors (see [Kim et al., 2021](#); [Gavilanes et al., 2018](#); [Barger et al., 2016](#); [Swani et al., 2013](#)). Of those, the medium itself (i.e. the platform) has been found to be an essential factor driving advertising response ([Voorveld et al., 2018](#); [Calder et al., 2009](#)), as well as other positive downstream behaviors that may be invisible ([Van Doorn et al., 2010](#)). Media engagement manifests across a variety of experiences ([Calder et al., 2009](#)) but is generally a good demonstration for predicting advertising effectiveness ([Davis Mersey et al., 2010](#)).

Engagement in a hierarchy of effects

The earliest studies of engagement conceived engagement along a range of outcomes on a low to high continuum. As the literature stream grew, similar continuums were conceived but were then segmented by consumption (clicks), contribution (commenting) and co-creation ([Muntinga et al., 2011](#); [Shao, 2009](#)). In several contexts, consumer-brand engagement has been examined in terms of brand outcomes relative to cognitive, affective and behavioral engagement ([Vander Shee et al., 2020](#)) among audiences. These hierarchies of effects have been upheld as a prevailing understanding of how advertising influences consumer decisions ([Smith and Swinyard, 1982](#)).

[Gavilanes et al. \(2018\)](#) conceptualize digital consumer engagement with advertising alongside these advertising processing effects. In their model, the authors extend much of the seminal work in engagement literature and connect the hierarchy of effects with advertising processing for a theoretical understanding of how Facebook metrics can be used to understand digital consumer engagement in social networks. For instance, the group of consumers who represent a potential brand advocate may be more influential as they actively advocate for a company by sharing their content ([Hollebeek, 2011](#)). Among the engagement effects, advertisers aim to accomplish a myriad of goals related to persuasion

(Vakratsas and Ambler, 1999). These outcomes can drive useful business intelligence regarding consumer insight (Stone and Woodcock, 2014).

Extending beyond the first hypothesis, which sought to show that native advertising placements in a social network will lead to improved engagement results when compared to banner advertisements, the second hypothesis seeks to show that exposure to native advertising will result in positive outcomes of engagement alongside the full hierarchy of effects conceptualized by Gavilanes and his colleagues using clicks, comments and shares.

- H2. Native advertising impressions will have a positive relationship with (H2a) clicks, (H2b) comments and (H2c) shares.

Sponsored social media posts

Native advertising scholarship has been widely accepted based on the belief that it is more effective than alternative placements (Marks *et al.*, 2019). As such, native advertising scholars tend to believe that varying motivational factors lead to differences among native advertising placement results. For instance, Lu *et al.* (2014) revealed that sponsorships on social media resulted in a positive relationship with purchase intention. However, of the three types of native advertising, Marks *et al.* (2019) suggest that social media appear to have more variation in the findings, which in turn suggests additional motivations that may vary between the browsing behaviors on these platforms. Lee *et al.* (2016) suggested that marketers can further delineate boundary conditions of engagement with specific audience groups, which could provide insight into how or why variation occurs.

Additionally, native advertising can benefit by being personalized for a user in their feed (Jung and Heo, 2019), but as some researchers have argued, effectiveness research to date has focused too heavily on short-term outcome metrics (e.g. Araujo *et al.*, 2020; Lambert, 2018; Lu *et al.*, 2014). Others argue that not all measures of consumer engagement are equal, especially within social media where brand engagement now includes two-way communication (Fulgoni, 2016). Moreover, the available data can help researchers understand the impact of social media advertising on attitude and purchase behavior (Marks *et al.*, 2019). As such, marketers may be able to see improved results in their downstream engagement depending on the objective of their advertisement and in turn by reaching a relevant consumer.

Engagement by advertising objective

An advertising objective is selected by an advertiser based on what the advertiser wants someone to do when a person sees the ad (Help: choose the right ad objective, n.d). In other words, it is posited that certain boundaries that are defined by the advertiser may perform better for native advertising placements than others. Specifically, these boundary conditions are operationalized based on an advertiser's given objective. Based on the widely adopted stream of native advertising literature, it is hypothesized that consumption (e.g. clicks, comments and shares) will be positively related to digital consumer engagement across the current sample of advertiser's objectives as shown in Figure 2.

Yet, because the Facebook feed is curated with posts from "friends" and provides a unique advertising experience for each user, users may be already engaged with a stream of content that has been highly personalized for them. Therefore, an opportunity remains to understand the circumstances where native advertising is more effective when the marketing and planning focus becomes engagement (Figure 3) (Araujo *et al.*, 2020).

Figure 2.
Advertising
processing model:
digital consumer
engagement in social
networks (Gavilanes
et al., 2018)

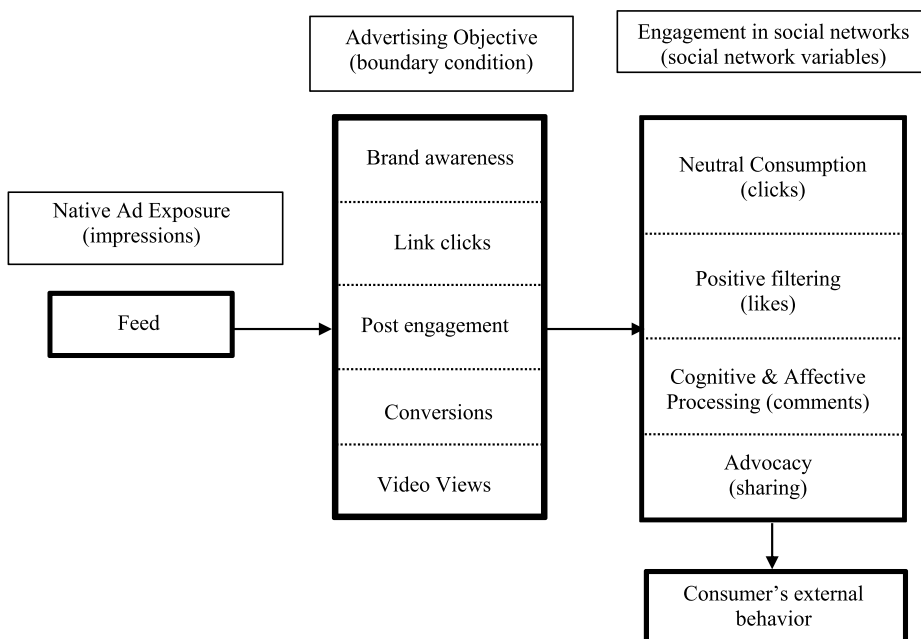
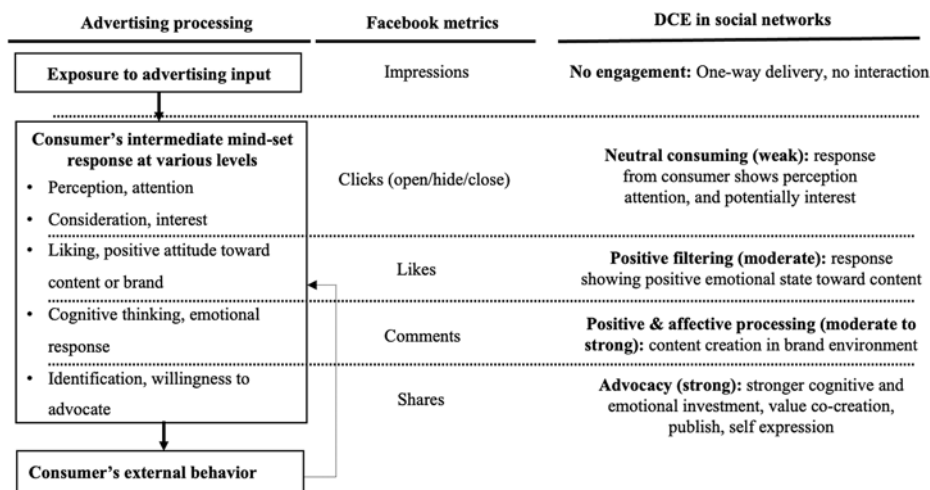


Figure 3.
Advertising
processing model:
digital consumer
engagement by
advertising objective
in social networks

In sum, the following hypotheses aim to empirically advance a more nuanced understanding of native advertising in a social network. The boundary conditions used to examine these effects are operationalized using the objectives that were employed by this sample of advertisers when scheduling their advertisements on Facebook.

- H3. For advertisers who chose the *brand awareness objective*, native advertising impressions will have a positive relationship with (H3a) clicks, (H3b) comments and (H3c) shares.
- H4. For advertisers who chose the *link clicks objective*, native advertising impressions will have a positive relationship with (H4a) clicks, (H4b) comments and (H4c) shares.
- H5. For advertisers who chose the *conversions objective*, native advertising impressions will have a positive relationship with (H5a) clicks, (H5b) comments and (H5c) shares.
- H6. For advertisers who chose the *post-engagement objective*, native advertising impressions will have a positive relationship with (H6a) clicks, (H6b) comments and (H6c) shares.
- H7. For advertisers who chose the *video views objective*, native advertising impressions will have a positive relationship with (H7a) clicks, (H7b) comments and (H7c) shares.

Method

Data

Using 26 months of Facebook advertising data, this study examines native advertising effects using a unique, real-world data set. The data set in its entirety included 20,711 advertising observations (e.g. ads) that were collected from a convenience sample of 10 advertisers that ran during the period from November 23, 2019, to December 22, 2021. The data include the full date range of historical data that was accessible on Facebook which was exported from the advertiser's Facebook advertising accounts (e.g. Facebook Ads Manager) before being shared for use in research. The data were then combined and aggregated in R. All identifiable data were removed prior to analysis.

Extending previous marketing literature, the authors examined advertising effectiveness across advertising processing variables that were measured as outcomes of digital consumer engagement with advertising in a social media network. The sample group of advertisements included 20,711 advertisements in a variety of placements, with 3,919 of them being native advertisements (e.g. feed) and 1,011 being banner advertisements (e.g. right hand column).

The sample of advertisements is presented by variables in Table 1. Each of the data was collected and measured using integral data. The five objectives that were used most by the advertisers were conversions ($n = 5,502$), link clicks ($n = 5,339$), post-engagement ($n = 2,544$), brand awareness ($n = 1,364$) and video views ($n = 2,160$). These five objectives were used to create the sample advertising data for the model results presented in Table 3. For the first analysis, the sample included both the native advertisements and the banner advertisements ($n = 4,930$), but for the remaining analyses, the sample consisted only of the native advertisements ($n = 3,919$).

All analyses were run in R, and the results are presented below after describing the variables and methods used for the analysis.

Facebook metrics: measuring digital consumer engagement in social networks

Online engagement can be measured using clicks, comments and shares or any interaction that a consumer has taken with a piece of content. For marketers who prioritize engagement as a measure of advertising effectiveness, it is useful to delineate between these.

Impressions. Impressions are a count measure of a message that was potentially seen by a social media user. Impressions are unique from reach and measure instead the number of

Table 1.
Descriptive statistics
of variables

Variables	Observation	<i>M</i>	Standard deviation	Minimum	Maximum
Dependent variables					
Link clicks	20,711	34.39	221.18	1	11,616
Post comments	20,711	8.23	45.36	1	966
Post shares	20,711	12.13	77.13	1	1,975
Variables					
Impressions	20,711				
Advertising objectives	<i>n</i> (%)				
App installs	2,981 (14.39)			0	1
Brand awareness	1,364 (6.59)			0	1
Conversions	5,502 (26.57)			0	1
Event responses	58 (0.28)			0	1
Link clicks	5,339 (25.78)			0	1
Messages	16 (0.08)			0	1
Multiple	602 (2.91)			0	1
Page likes	32 (0.15)			0	1
Post-engagement	2,544 (12.28)			0	1
Reach	113 (0.55)			0	1
Video views	2,160 (10.43)			0	1

times the message was seen versus reached, which measures the number of people that the advertisement reached.

Link clicks. This engagement variable measures the concept of consumption and has been used to understand purchase intent. In this study, link clicks are used to measure the number of clicks on links within an advertisement that lead to advertiser-specified destinations, on or off Facebook. Facebook also reports links, which are a measure of any click that occurs on the post. Link clicks are conceived as being more relevant for this study as they suggest interest and intent.

Post comments. Post comments are the second metric of engagement and suggest a higher level of advertising processing in the hierarchy of effects in Figure 1 and are measured when a user comments on the advertisement.

Post shares. The post shares variable is the third engagement metric of interest and represents the fourth level of digital consumer engagement with advertising and suggests the highest level of engagement. Post shares are a measure of advocacy and are a count of the number of times an advertisement was shared.

Differences in native advertising compared to banner advertising

To examine the first hypothesis and understand if a statistically significant difference in advertising effectiveness existed between the native and banner advertisements, a *t*-test was used to dichotomously compare the banner ads with the feed ads. Previous marketing research has used similar tests to understand the marketing impact of two conditions (Zhang and Patrick, 2021). Because banner advertisements cannot be shared or commented on, differences were compared based on their condition (feed or banner ad) and then their impact on dependent measures of impressions as well as link clicks was measured. Results with a small *t*-value indicated that more similarity existed between the samples, whereas a large *t*-score will indicate a larger difference between the groups.

One-way fixed effects model

Panel data, which are also called longitudinal or cross-sectional time-series data, are becoming increasingly available through social networks. Panel data tend to be good at

accounting for heterogeneity (Erdem, 1996) and are especially useful for modeling (Park, 2011). Many of the issues that can occur in regular ordinary least squares regression (OLS) can be eliminated by using panel data analysis which aggregates variables and considers serial correlation of observations to eliminate biased estimates. Panels have been widely used in market research (Vomberg and Weis, 2022; Lohse *et al.*, 2000) due to their informative nature, increased variability, low collinearity and increased degrees of freedom (Baltagi, 2001). To be classified as panel data, observations measure the same units over several time periods (Kennedy, 2008) and can be defined by some type of unit like location (e.g. state, county), companies, individuals or, in this case, advertisers.

For each model, data were aggregated by advertiser and time. Each of the regressions used month data and integers, or count data, of impressions, clicks, comments and shares. The first group of models (H2a–H2c) regressed exposure to native advertising on link clicks, post shares and post comments. The regression models with significant variables are presented in Table 2. The second group of models (H3) explored the same native advertising exposure effects but segmented the data by advertising objective and regressed exposure to native advertising on link clicks, post shares and post comments. The regression models with significant variables are presented in Table 3.

In *R*, with the function “pFtest()”, the data was tested for fixed effects (FEs), where the null hypothesis is that pooled OLS is better than FEs. In all cases, the null hypothesis was rejected in favor of the alternative that there were significant FEs. Thus, a one-way FE model was employed for each regression. In the FE model, it is assumed that if unobserved characteristics do not change over time, dependent variable changes must instead be related to FEs. This FE model is well suited for studying effects across the panel of advertisers. In an FE model, it is assumed that there are unobserved influences (e.g. content or emotion) which are controlled for which can make FE models good for investigating causal relationships between variables.

Results

Overall, the results suggest that native advertising was significantly related to outcomes of engagement across the hierarchy of effects of digital consumer engagement with advertising. Advertising consumption effects of impressions and clicks were revealed to be significantly higher in native advertising compared to banner advertising. Native advertising was also found to have a statistically significant relationship with digital consumer engagement, resulting in consumption (e.g. clicks), cognitive and affective processing (e.g. comments) and advocacy (e.g. sharing).

On examining H1a, the hypothesis was supported. A *t*-test with impressions as the dependent variable revealed that the 3,919 advertisements in the feed condition ($M_{\text{native}} = 9,615.33$, $SD = 4,1767.30$) generated more impressions than the 1,011 advertisements using banner advertisements ($M_{\text{banner}} = 3,784.69$, $SD = 3,7795.84$) and that the difference was significant, $t(1,717) = 4.250$, $p < 0.001$.

Table 2.
Native advertising
effects by digital
consumer engagement
with advertising

Predictor	(1) Link clicks	(2) Comments	(3) Shares
Impressions	<0.001***	0.002**	0.038*
<i>t</i> -Statistic	7.960	3.2	2.11
<i>N</i>	121	121	121
<i>R</i> ² / <i>R</i> ² adjusted	0.536/0.483	0.394/0.339	0.334/0.273
Notes: * <i>p</i> < 0.05, ** <i>p</i> < 0.01, *** <i>p</i> < 0.001			

					Digital consumer engagement
Advertiser objectives	Predictor	(1) Link clicks	(2) Comments	(3) Shares	
Brand awareness	Impressions	<0.001***	<0.001***	<0.001***	
	<i>t</i> -Statistic	16.97	8.26	8.4	
	<i>N</i>	18	18	18	
	<i>R</i> ² / <i>R</i> ² adjusted	0.960/0.943	0.850/0.788	0.8546/0.794	
Link clicks	Impressions	<0.001***	<0.001***	<0.001***	
	<i>t</i> -Statistic	21.76	20.36	16.81	
	<i>N</i>	63	63	63	
	<i>R</i> ² / <i>R</i> ² adjusted	0.896/0.883	0.883/0.868	0.837/0.816	
Conversions	Impressions	<0.001***	<0.001***	0.001**	
	<i>t</i> -Statistic	14.71	5.32	4.03	
	<i>N</i>	21	21	21	
	<i>R</i> ² / <i>R</i> ² adjusted	0.935/0.914	0.653/0.53	0.512/0.360	
Post-engagement	Impressions	0.66	<0.001***	<0.001***	
	<i>t</i> -Statistic	−0.44	14.29	13.240	
	<i>N</i>	50	50	50	
	<i>R</i> ² / <i>R</i> ² adjusted	0.004/−0.134	0.826/0.802	0.803/0.775	
Video views	Impressions	<0.001***	<0.001***	<0.001***	
	<i>t</i> -Statistic	7.060	12.85	13.32	
	<i>N</i>	18	18	18	
	<i>R</i> ² / <i>R</i> ² adjusted	0.819/0.721	0.938/0.903	0.942/0.910	
Notes: * <i>p</i> < 0.05, ** <i>p</i> < 0.01, *** <i>p</i> < 0.001					

Table 3.
Digital consumer engagement with advertising by advertising objective

H1b was also supported, which compared differences in advertising consumption between the two groups and found a statistically significant improvement in the click results from native advertising ($M_{\text{native}} = 67.54$, $SD = 368.23$) compared to clicks from banner advertisements ($M_{\text{banner}} = 16.41$, $SD = 77.73$), $t(1,698.20) = 5.62$, $p < 0.001$.

H2a–H2c proposed that exposure to native advertisements would be significantly related to digital consumer engagement with advertising and have positive relationships with a hierarchy of advertising effects including clicks, comments and shares. The results in Table 2 show that H2a–H2c were supported and that impressions were positively related and statistically significant in all three models. In other words, native advertising exposure was significantly related to increases in the number of link clicks, comments and shares. The models explained between 27 and 48 per cent of the given predictor variance for the regression models.

The final group of hypotheses (H3a–H7c) sought to measure the relationship between impressions by advertising objective and clicks, comments and shares. As suggested in the literature, these were found to be partially supported. The results showed that in 14 out of 15 models, exposure to native advertising was positively related to increases in the digital consumer engagement variables of interest: link clicks, comments and shares. However, for advertisers who chose post-engagement for their advertising objective (H6a), impressions were not statistically related to increases in link clicks.

The models presented in Table 3 explained between 0 and 94 per cent of the given predictor variance for the regression models. The results of the models are discussed below.

Discussion

The social media environment, compared to traditional media like television, supports the use of native advertising to build the consumer–brand relationship. To examine this, this study investigated native advertising occurring on Facebook. After conceptualizing the

relationship between advertising processing and increased engagement in social networks, the authors showed that consumption effects were significantly better for native advertising when compared to banner advertisements. Overall, the study showed that exposure (e.g. impressions) to native advertising had a significant relationship with advertising processing, consumption and advocacy.

Ads that reach consumers through the Facebook network while natively consuming content in their feed resulted in the most consistent and statistically significant engagement results. The findings suggest that native ads in the feed are highly effective at driving engagement.

The results provide evidence that marketers using native advertising can expect positive results in digital consumer engagement with advertising.

Consistent with what [Jung and Heo \(2019\)](#) found, the first hypotheses predicted that native advertisements would be more effective than banner advertisements on Facebook. In support of this, the hypotheses were also confirmed true, and when compared to banner advertisements, native advertisements were not only found to be seen by more people but also engaged with by more people. These findings were in line with the expected outcomes, but with the limited number of features available to a Facebook user seeing a banner advertisement, engagement with a banner advertisement can only be measured through exposure and link clicks. Although unlikely, there is some chance for an outlying situation to occur where a banner advertisement leads to advocacy or word of mouth on behalf of a brand.

The second hypothesis sought to expand native advertising more broadly by integrating digital consumer engagement with the advertising model. Notably, this analysis used advertising data aggregated by advertiser and month, whereas the third hypothesis analyzed advertising data aggregated by advertising objective. Each of the models in [H2a–H2c](#) showed that exposure to native advertising had a significant relationship with the dependent measures of consumer engagement. Interestingly, the first model reported the highest explained variance ($R^2 = 0.483$) and suggested that in the aggregate, exposure to native advertising was a strong predictor of link clicks.

The results, presented in [Table 2](#), could be deceiving without further investigation as they seemingly suggest that advertisers who use native advertising in social media can expect users to click on the content. However, the third hypothesis allowed a more granular measure of these effects which were explored in the extension of the work by [Lee et al. \(2016\)](#) that suggested that certain boundary conditions may help marketers with specific audience groups. Recall also that Gavilanes and his colleagues had suggested the fourth dimension of engagement to be representative of a more valuable consumer. Extending from these findings, the study sought to understand how exposure to native advertising may vary across groups, as defined by the advertising objectives chosen by the sample that may lead to varying outcomes of consumer engagement. This research evaluated native advertising placements, then, by the five objectives that were used most by this sample: brand awareness, link clicks, conversion, post-engagement or video views as their advertisement objective.

Interestingly, exposure to native advertising for the advertisers who chose brand awareness for their ads found that exposure explained 94 per cent of the link click predictor variance for the model. Of the models used in the study, this was the highest explained variance and is especially interesting given that the advertising objective of awareness is usually related to reaching an audience. On the other hand, the models that reported the highest explained variance for comments and shares were for those advertisers who chose video views as their objective.

In contrast to the findings from [H2a–H2c](#), clicks were not found to be positively and significantly related to all native advertising exposure. Specifically, for advertisers who chose post-engagement as their advertising objective, exposure to native advertising was

not significantly related to increases in link clicks (H6a). Conceptually, clicks are lower on the hierarchy of engagement. Thus, for advertisers choosing the engagement objective, it is possible that Facebook is prioritizing the advertisements for people most likely to comment or share. This insight is especially useful for interactive marketers who may be measuring outcomes in aggregate or who look at the measure of engagement in aggregate, which would calculate a sum for clicks, comments and shares. The results suggest that advertisers who are seeking traffic should instead use brand awareness, conversions, link clicks, or video views for their objective. Similarly, Hollebeek and Gavilanes and their colleagues had conceptualized the fourth level of engagement to be unique from other measures of engagement and argued that shares are representative of brand advocacy and yield different implications. The models used in the second hypothesis (H2) examined shares as the dependent variable of interest and explained about 27 per cent of the variance.

Turning toward the second group of models presented in Table 3, the authors saw strong support for four out of five models which were segmented by advertising objective. Of the five models, the model that included native advertisers who had chosen to use conversions for their advertising objective performed poorly and offered a new lens through which to examine native advertising effectiveness by advertising objective.

Theoretical contribution

The results support the findings by Campbell and Marks (2015) who had suggested that the positive effects related to native advertising will perform better in the long run. This contradicts other native advertising literature that had suggested that sponsored posts in a social media feed could result in negative effects related to a person's knowledge or awareness of the persuasion tactics used by the advertiser to reach them (Kim *et al.*, 2019; Taylor, 2017; Wojdyski and Evans, 2016; Boerman *et al.*, 2014). Instead, the current study shows that native advertising placements have a positive and significant relationship with digital consumer engagement outcomes and that native advertisements drive significantly more of the desired engagements when compared to banner advertisement placements. Thus, outcomes of coproducing or cocreating suggest that consumers are deeply involved with the message. In turn, native advertising offers marketers the chance to build a positive, long-lasting and loyal relationship with consumers. Conceptualizing the engagement behaviors alongside low to high levels of advertising processing, the findings offer insight into how these measurements relate to the positive benefits afforded to marketers using native advertising in a social network.

In an extension of the literature, the engagement levels represent unique ad processing levels or states occurring at different relationship phases between a consumer and a company (Hollebeek, 2011). Because researchers have used a variety of definitions of engagement and could benefit from some consistency in defining engagement (Barger *et al.*, 2016), the model offers another examination of how native advertising effectiveness can be conceptualized through engagement outcomes. Marketing researchers tend to agree that when conceptualizing engagement, the measures should be multidimensional (Voorveld *et al.*, 2018; Hollebeek *et al.*, 2014). Thus, having operationalized outcomes of digital consumer engagement alongside advertising, the findings can help close the gap between practice and research (Lamberton and Stephen, 2016) and contribute to an understanding of the role of native advertising in digital consumer engagement by advertising objective. By delineating native advertisements by boundary conditions, the native advertising literature is enriched, contributing to research exploring why sponsored posts should be a marketing priority.

Managerial implications

Interactive marketing has brought about many new opportunities for organic consumer-brand relationships to occur, and advertising can act as an antecedent of engagement

(Gavilanes *et al.*, 2018). Marketers can use these results to understand and strategically conceptualize how low to high levels of engagement variations occur in a social network. Facebook offers incredible opportunities to market products and services, but the access to native advertising and potentially related outcomes of engagement differentiate social network advertising from other forms of sponsored placements. The findings provide another lens for marketers to consider when strategically allocating their media placements in social media by placement and objective. Advertisers should consider their intended goals for their audience and the role an individual campaign objective may play and address personalization in their advertising content strategy accordingly.

Within the Facebook advertising ecosystem, an advertiser may reach users on Facebook or Instagram, as well as other channels and placements owned by Facebook. The placements offer several targeted options between the device a person is using, either mobile or desktop, or any of the several placements available through the network. Placement options may either be automatically selected by default (e.g. run of site inventory) or specified by the advertiser and scheduled specifically on a platform, device or placement. When automated, the placements are scattered across the advertising ecosystem as suggested by Facebook, delivered to several locations on Facebook, Messenger and Instagram. Advertisers using the Facebook Ad platform are prompted by Facebook to use the automated placements, suggesting that “targeting works the same across all available placements, so including more placements often helps you find a wider audience” (Facebook, 2021). However, the results show that native advertising leads to increased consumer engagement and, in turn, suggest that when creating and placing media, Facebook advertisers should disable the automation of advertisement placements in these media ecosystems to improve engagement.

Limitations and future research

This study empirically analyzed the advertising effectiveness of native advertising within the Facebook advertising ecosystem. The results should be considered bearing in mind some limitations. First, behavioral trace data has been gaining ground in the social sciences (Wu and Taneja, 2021) and is the source for this study. These behavioral data are commonly referred to as “digital traces” and are an important criterion relating to the study of computational communication science (van Atteveldt and Peng, 2018). The validity of these behavioral trace data as measures of advertising processing is still being accessed. Social media data helps explain consumer behavior (Voorveld, 2019), but it does not tell us why those clicks, comments and shares occurred. Therefore, future studies may use validity tests to extend support in understanding the relationship between digital consumer engagement and related outcomes of cognitive and affective processing of advertisements.

Similarly, relying on behavioral trace data provides its own set of issues relating to platform episteme (Wu and Taneja, 2021). One example of this limitation in the current study lies in the model’s use of “likes” as a measure of advertising processing in social networks. While real-world data gives us insight into the ad processing that is occurring through measures of consumer behavior, researchers remain limited by the data provided by the platform. In this instance, “likes” are no longer individually reported but are instead grouped as “reactions” and include a variety of positive and negative sentiments of emotion. Therefore, this study excluded the reactions. Future research should explore how these sentiments relate to engagement.

Personalization, critical mass, access and risk, along with technical factors relating to the platforms, can impact engagement and consumption (Di Gangi and Wasko, 2016). For instance, banner ads are visible to desktop users, whereas they do not appear within the mobile Facebook app. Thus, mobile devices may be more conducive to native advertising

than desktop experiences where banner advertisements generally appear. Because this data set did not include measures of where the feed or banner ads reached people (e.g. on mobile versus desktop), it should be noted that most people use the Facebook mobile application where banner ads are not displayed. However, there is no way to know exactly how many of the ads were delivered via mobile versus desktop and vice versa. Future studies may work to further explicate this.

Conclusion

Native advertisements reaching consumers in their feed offer a promising opportunity for marketers seeking to build consumer–brand relationships. Perhaps, the single most attractive feature for a social media advertiser is the platform’s ability to encourage a consumer–brand relationship through targeted engagement. Additionally, social media ad spending in the US is projected to achieve a market volume of \$51m by 2024 (Kunst, 2022). This may be in part attributed to the expansive number of communication activities available to a social media user, but it is also likely related to the engagement benefits afforded to an advertiser placing advertisements in the feed in a social network. Determining the appropriate mix of reach and frequency is a challenge for marketers (Araujo *et al.*, 2020). Therefore, native advertising may become the priority placement for delivering advertisements in a social network.

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